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Department of Computer Science

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Final Internship Report:

New York University-

eBrain Lab

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Executive Summary

In this report, I will provide an overview of my summer internship at the eBrain Lab at New York University in Abu Dhabi, under the supervision of Dr Mohammed Shafique and the mentorship of Abdul Basit.

The project I worked on during the internship was the classification of electroencephalography (EEG) data from the BCI Competition 2008 – Graz data set A, which contained motor imagery (MI) of 9 subjects' left and right hands, feet, and tongue. My role was to explore the dataset, develop machine learning models to do the classification task, and test the robustness of mine and a state of the art model under different conditions.

Over the eight weeks, I studied the common EEG model popular in the field and decided to use EEGNet, I also reviewed what I know of Pytorch and gained new knowledge about the library and other new libraries used for EEG data. I explored different data augmentation models and arrived at many conclusions and explanations for model behaviour.

This internship notably enhanced my programming skills and especially in Pytorch, and the library primarily used in courses at Khalifa University was TensorFlow. I practiced and enhanced my skills in teamwork, presentation, and data analysis through the meetings and discussions I held with my mentors.

The Organization's Business Sector

The internship took place in a higher education sector, within the field of human computer interaction, specifically brain computer interaction (BCI). BCI systems use brain signals to execute commands for output devices, where the signals are analyzed and translated by the system's program [1]. This field's main goal is to restore function to people disabled by injuries or medical conditions [1]. Recent advancements in the field include rehabilitation and neurofeedback, fatigue and stress detection, motor imagery for neuroprosthetics, and virtual reality applications [2]. Improved EEG processing, machine learning models, and wearable sensors have increased accuracy and usability [2]. Although challenges have been shown to appear in handling noisy, non-stationary signals, and training users to effectively use the technology [2].

Overview of the organization

My internship was conducted at the eBRAIN Lab, which is part of the Division of Engineering at NYU Abu Dhabi. The laboratory is directed by Dr Muhammad Shafique, Dr of Computer Engineering at NYUAD. The eBRAIN Lab focuses on the research and development of advanced machine learning technologies, secure and reliable computing systems, and the implementation of artificial intelligence models in embedded and real-world applications [3].

The lab also investigates the application of machine learning in areas such as smart healthcare, robotics, autonomous systems, agriculture, and industrial applications [3], [4]. Its research includes the development of intelligent and efficient computing systems and the application of machine learning techniques to a variety of problems.

Internship Program

My internship was conducted in the eBrain Lab at NYU Abu Dhabi, specifically within the EEG based motor imagery classification project. The internship began on the 1st of June and concluded on the 24th of July. During my internship, I worked on a project involving the analysis and classification of EEG data. The main purpose of the project was to study how different machine learning models perform when EEG signals are affected by changes such as noise, missing data, scaling, missing time segments, and missing frequency information.

At the beginning of the internship, I familiarized myself with the BCI Competition 2008 Graz Dataset A. The dataset contained EEG recordings from nine subjects performing four motor imagery tasks. I learned how the data was structured and used the MOABB library to load, organize, and preprocess the recordings. I also divided the subjects into training, validation, and testing groups and tested multiple evaluation methods, including cross validation.

I then implemented and compared several EEG classification models. These include a simple neural network, EEGNet, and a Common Spatial Pattern with Linear Discriminant Analysis model. Working with these models allowed me to gain practical experience using both PyTorch and Keras. I also learned how differences in framework settings, layer order, default parameters, and software environments can affect model implementation and performance.

A major part of my internship involved preparing experiments to evaluate model robustness. I modified the EEG data by adding noise, scaling selected channels, removing channel information, removing portions of the signal, and filtering frequency ranges. I tested whether these changes were applied to the training data, testing data, or both.

During the final stage of the internship, I reviewed model performance for individual subjects and classes. I examined confusion patterns, class recall, differences between subjects,

correlations between recording sessions, and the frequency of model predictions. I also used statistical comparisons to identify meaningful differences between motor imagery classes.

Throughout the internship, I documented my weekly progress, organized experimental results, prepared visual summaries, and discussed challenges. The program gave me practical experience in EEG signal processing, machine learning, data analysis, and research methodology. It also improved my problem solving skills, technical independence, and ability to interpret and communicate experimental work clearly.

Learning experiences

The internship gave me practical experience with EEG data structures, motor imagery classes, channel based signal representations, and the difficulties of learning from noisy time series. I became more comfortable reading data, checking tensor dimensions, and verifying that transformation application onto subjects, channels, and time windows.

I learned to use MOABB for loading and organizing EEG datasets and for constructing the CSP + LDA benchmark. I also strengthened my PyTorch skills by building a simple convolutional model and converting EEGNet from Keras.

The project also developed my understanding of experimental controls and project versioning. Applying the same test conditions to both training and test data, and what findings we can conclude and questions we can answer under different scenarios. The first condition measures performance on consistently transformed data, while the second measures sensitivity to distribution shift. The large differences observed between these conditions showed why preprocessing must be documented and reproduced exactly at deployment time.

I also gained experience interpreting confusion matrices, recall, per class behavior, per subject trends, pairwise comparisons, and correlation across sessions. These analyses helped move the project beyond model ranking and toward an explanation of where and why the models failed.

The main early challenge I faced was understanding and manipulating the dataset. I relied on MOABB documentation, guides, and tutorials to confirm how subjects, trials, channels, and labels were organized. I repeatedly inspected shapes and outputs to ensure that the correct dimensions were being transformed.

A second challenge was reproducing EEGNet across Keras and PyTorch. Differences in defaults and layer ordering caused significant accuracy changes, and the original implementation required a new, separate environment. Fixing these issues improved my ability to compare codes and manage technical dependencies.

The project heavily depended on careful verification, and clear documentation to ensure that the findings were correct and comparable to each other. As research results can be misleading when evaluation splits are biased, when preprocessing is inconsistent, or when configuration changes are not recorded.

Conclusion

The internship provided hands on experience in EEG analysis, machine learning implementation, robustness testing, and research communication. The results I concluded were a variety of findings regarding the behavior of models, when training and test preprocessing were consistent, several disturbances caused only moderate accuracy decrease. When a transformation was applied to only one partition, accuracy sometimes collapsed. The effect was especially clear for scaling and noise.

The class and subject analyses also showed that performance was not uniform. Feet imagery was particularly sensitive and was often confused with tongue imagery. Hand imagery was comparatively robust but left and right hands remained difficult to separate. Several subjects were consistently challenging, while others appeared difficult mainly for EEGNet. Overall, the project showed that robustness, reproducibility, and subject level analysis are critical to meaningful EEG classification research.

References

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Appendix

One of the labs that are used by the eBrain lab

